

Question 1:

Method	Stack Contents	Return Value
push(5)	(5)	
push(3)	(5, 3)	
pop()	(5)	3
push(2)	(5, 3)	
push(8)	(5, 2, 8)	
pop()	(5, 2)	8
pop()	(5)	2
push(9)	(5, 9)	
push(1)	(5, 9, 1)	
pop()	(5, 9)	1
push(7)	(5, 9, 7)	
push(6)	(5, 9, 7, 6)	
pop()	(5, 9, 7)	6
pop()	(5, 9)	7
push(4)	(5, 9, 4)	
pop()	(5, 9)	4
pop()	(5)	9

Question 2:

Method	Queue Contents	Return Value
enqueue(5)	(5)	
enqueue(3)	(5, 3)	
dequeue()	(3)	5
enqueue(2)	(3, 2)	
enqueue(8)	(3, 2, 8)	
dequeue()	(2, 8)	3
dequeue()	(8)	2
enqueue(9)	(8, 9)	
enqueue(1)	(8, 9, 1)	
dequeue()	(9, 1)	8
enqueue(7)	(9, 1, 7)	
enqueue(6)	(9, 1, 7, 6)	
dequeue()	(1, 7, 6)	9
dequeue()	(7, 6)	1
enqueue(4)	(7, 6, 4)	
dequeue()	(6, 4)	7
dequeue()	(4)	6

Question 3:

```
import java.util.ArrayDeque;

public class ArrayDequeDemo{
    public static void main(String[] args) {
        ArrayDeque<Pair> a = new ArrayDeque<>();

        Pair<String, Integer> P1 = new Pair<>("Books", 4);
        Pair<String, Integer> P2 = new Pair<>("Mobiles", 3);
        Pair<String, Integer> P3 = new Pair<>("Laptops", 2);
        Pair<String, Integer> P4 = new Pair<>("Boards", 5);

        a.add(P1);
        a.addFirst(P2);
        a.addFirst(P3);
        a.addLast(P4);

        System.out.println("Then contents of Deque are: " + a);
        //The contents of Deque are: [(Laptops,2), (Mobiles,3), (Books,4),
        (Boards,5)]
    }
}

class Pair<A,B> {
    A first;
    B second;
    public Pair(A a, B b) { // constructor
        first = a;
        second = b;
    }
    public A getFirst( ) { return first; }
    public B getSecond( ) { return second;}

    public String toString(){
        return "(" + getFirst() + "," + getSecond() + ")";
    }
}
```

Output: Then contents of Deque are: [(Laptops,2), (Mobiles,3), (Books,4), (Boards,5)]